

Design of Wall Art Utilizing Dynamic Color Changes through Photoelasticity

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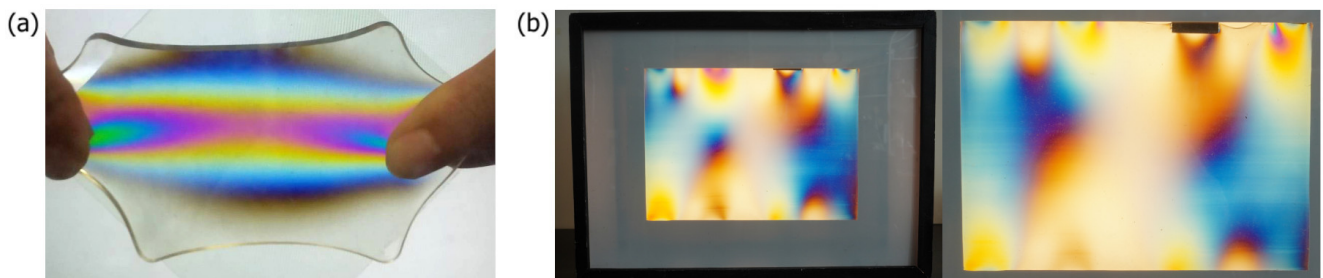


Figure 1: (a) General photoelastic phenomenon. (b) Color change effect created by our prototype.

Abstract

This research proposes a dynamic wall art piece that utilizes color changes through photoelasticity. To achieve this effect, we have developed a device that stretches a single sheet of gel material from multiple points. This device allows for gentle, organic color transitions that cannot be replicated by pixel-based displays like LCDs, offering a unique aesthetic experience as wall art.

ACM Reference Format:

Ryota Nakayama, Soshi Takeda, Gakuto Sekine, and Yuichiro Katsumoto. 2024. Design of Wall Art Utilizing Dynamic Color Changes through Photoelasticity. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681756.3697902>

1 Introduction

In this research, we designed a wall art piece that utilizes dynamic color changes through photoelasticity. This phenomenon occurs when polarized light passes through a transparent, flexible material, causing color changes due to material deformation (Figure 1a). Unlike pixel-based displays such as LED matrices and Liquid Crystal Displays (LCDs), photoelasticity produces gentle, analog color changes with subtle nuances. We developed a device that can

dynamically change colors and patterns by adjusting the points where the material is stretched and the amount of force applied. This device is intended to be displayed as wall art in hotels, cafes, and public spaces.

2 Related Works

The objective of this research is to develop a technique for generating various colors by deforming materials and applying this method to wall art. One example of a material that changes color through deformation is Photonic Rubber[Fudouzi and Sawada 2006]. In its normal state, this material has a reddish tint, but when stretched, the elongated areas turn green. This material is made of polystyrene, and the deformation of its structure alters the wavelengths of light, producing visible colors. Another example using polarizing plates is Polagons[Sethapakdi et al. 2023], a polarizing mosaic structure that changes color depending on the direction of incident light, allowing users to control color changes. This structure consists of cellophane material and polarizing plates, and the color changes can be adjusted by modifying the angle of incident light. An example of applying photoelasticity as an input device is PhotoelasticTouch[Sato et al. 2009]. This transparent elastic interface leverages polarized light from an LCD display and the photoelastic properties of a transparent elastic material placed on the display to detect user interactions through material deformation.

The contribution of this research lies in enabling control of color changes through the photoelastic effect using a unique mechanism to stretch flexible materials, specifically designed for wall art.

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SA Posters '24, December 03–06, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-1138-1/24/12

<https://doi.org/10.1145/3681756.3697902>

3 Design

3.1 Concept

The device developed in this research utilizes the photoelastic effect. The photoelastic effect is a phenomenon in which flexible materials, such as rubber and silicon, appear to change color due to the filtering of light wavelengths when overlaid on a polarizing plate. A polarizing plate allows only light polarized in a specific direction to pass through. The principle of color extraction is based on the birefringence caused by the interaction of visible light with the polymeric crystalline structures formed as the flexible material is stretched, interacting with the polarizing plate (Figure 2). These crystals are commonly found in plastic products and gel sheet materials, and the colors change when the polarizing plates are overlaid on a stretched material and the angle is adjusted.

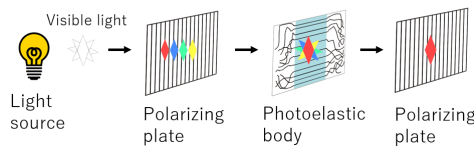


Figure 2: Processes leading to discoloration through photoelastic effects.

By exploiting this phenomenon, we aimed to design wall art capable of expressing analog gradations, which are difficult to achieve with colored LED light sources, and developed a mechanism using a high-torque stepper motor to stretch flexible materials at multiple points. The material chosen was a soft urethane sheet with a hardness of Asker C30, sold as "Hyper Gel Sheet" by Exseal Corporation. After testing different hardness levels, we found that C30 was the most suitable. Softer materials required more stretching to achieve visible color changes, while harder materials needed more force than the motor could provide. We experimented with stretching the material in multiple directions, but this prevented the formation of the crystalline regions necessary for color change. As a result, the final prototype was designed to stretch the material only in the vertical direction.

3.2 Implementation

The mechanism developed in this research utilizes a urethane sheet, measuring 410 mm in length, 280 mm in width, and 3 mm in thickness, which is stretched at eight specific points along both the top and bottom edges. This urethane sheet is securely held in place by a belt-driven device that is connected to a stepper motor, which can stretch the sheet by up to 40 mm, as demonstrated in Figure 3.

The overall dimensions of the wall art structure are 600 mm in width, 450 mm in length, and 250 mm in height. This structure features a polarizing plate, with the urethane sheet suspended between two additional polarizing plates. A white plate is integrated into the second polarizing plate, serving the purpose of neutralizing the light emitted from the light source, ensuring a balanced and uniform light distribution. A motor positioned on the right side pulls the sheet via the belt system, and the entire structure is securely fastened on all sides to prevent any displacement or misalignment during operation.

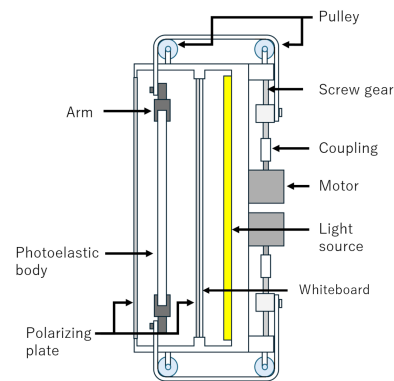


Figure 3: Our prototype structure diagram.

A stepper motor with a holding torque of 40 Ncm is employed in this system. The motor's shaft is connected to an M6 bolt through a coupling, which then holds the belt in place. As the motor rotates, the belt moves vertically, effectively pulling the urethane sheet to stretch it. The LED light source is softened by passing through the white plate and then proceeds through the polarizing plates and the urethane sheet. As the urethane sheet is stretched, birefringence occurs, leading to noticeable changes in color. The two polarizing plates are arranged diagonally to maximize the degree of color change, which is dependent on the angle at which the sheet is stretched and the tension applied.

An Arduino system is used to precisely control the motor and adjust the tension applied to the sheet during operation. In the tests conducted, it was found that stretching the urethane sheet with a force of 19.6 N in one direction produced vivid tints of orange, red, blue, and yellow, as illustrated in Figure 1b.

4 Conclusion

We developed a device for creating wall art using photoelasticity, which gently changes color through a gel material, polarizer, and eight actuators. Future work will focus on optimizing color vibrancy and specific tints, as well as enabling standalone operation. By rotating the gel sheet and filters or controlling the light source, we aim to achieve more complex color variations. We also plan to explore other applications, such as drawing letters with the altered hues or using color for information display.

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